Budget-Constrained Periodic Data Collection for Wildlife Monitoring: A Primal Dual Approach

Appendix A
Proofs of Theorems

Theorem 1: The solution $\boldsymbol{x}$ output by the algorithm is feasible for the primal problem, and the solutions $\boldsymbol{y}$ and z make the dual problem feasible.

Proof: When the algorithm terminates, all PoIs have completed periodic coverage, satisfying constraint $\sum_{n=1}^{N}\sum_{\tau=t}^{t+\Delta}(\mathbf{P}_n^{m\tau}\cdot x_n)\geq 1$ of the primal problem. The selected trajectories satisfy the budget constraint, thus satisfying $C\cdot\sum_{n=1}^{N}x_n\leq B$. At the same time, only those tight vehicle trajectories are selected and their variables are set to $x_n{=}\,1$, satisfying the integer programming requirement; therefore, the solution is feasible for the primal problem. Furthermore, under each fixed value of z, the dual variable $\boldsymbol{y}$ increases until some vehicle trajectories become tight, and the corresponding dual variables $\boldsymbol{y}$ are frozen. This ensures that for the selected vehicle trajectories $\mathbf{P}_n$, their trajectories satisfy $\sum_{t=1}^{T-\Delta}\sum_{m=1}^{M}(\sum_{\tau=t}^{t+\Delta}\mathbf{P}_n^{m\tau}\cdot y_{mt}) = C(1+z)$, while for the unselected vehicles, their trajectories satisfy $\sum_{t=1}^{T-\Delta}\sum_{m=1}^{M}(\sum_{\tau=t}^{t+\Delta}\mathbf{P}_n^{m\tau}\cdot y_{mt}) < C(1+z)$. Therefore, no vehicle trajectory exceeds the constraint range, and the dual problem is feasible.

Theorem 2: The cost $C(\widehat{\mathcal{P}})$ of the vehicle trajectory set $\widehat{\mathcal{P}}$ selected by the algorithm has an approximation ratio of $\Delta\cdot f$ with respect to the optimal solution $C(OPT)$ of the original problem. Here, f denotes the maximum number of times that any PoI m, $\forall t=1,2,...,T-\Delta, \forall m=1,2,...,M$, is covered at any time instant t by the vehicle trajectory set $\mathcal{P}$.

Proof:

$$\begin{aligned}\sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}C(\mathbf{P}_n) &= \sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}\sum_{t=1}^{T-\Delta}\sum_{m=1}^{M}\left(\sum_{\tau=t}^{t+\Delta}\mathbf{P}_n^{m\tau}\cdot y_{mt}\right)-\sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}C\cdot z\\ &= \sum_{t=1}^{T-\Delta}\sum_{m=1}^{M}\left(y_{mt}\sum_{\tau=t}^{t+\Delta}\sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}\mathbf{P}_n^{m\tau}\right)-\sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}C\cdot z\\ &\leq \sum_{t=1}^{T-\Delta}\sum_{m=1}^{M}\left(y_{mt}\sum_{\tau=t}^{t+\Delta}\sum_{\mathbf{P}_n\in\mathcal{P}}\mathbf{P}_n^{m\tau}\right)-\sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}C\cdot z.\end{aligned} \tag{1}$$

where the first equality follows from the primal–dual complementary slackness theorem, the second equality is obtained by exchanging the order of the parameters, and the first inequality holds because $\widehat{\mathcal{P}}\subseteq\mathcal{P}$, and thus

$$\begin{aligned}\sum_{\mathbf{P}_n\in\mathcal{P}}\mathbf{P}_n^{m\tau} &\leq \max\left\{\sum_{\mathbf{P}_n\in\mathcal{P}}\mathbf{P}_n^{m\tau},\,\forall\tau=1,2,\ldots,T-\Delta,\,\forall m=1,\ldots,M\right\}\\ &= f.\end{aligned} \tag{2}$$

therefore

$$\sum_{\tau=t}^{t+\Delta}\sum_{\mathbf{P}_\mathrm{n}\in\mathcal{P}}\mathrm{P}_\mathrm{n}^{\mathrm{m}\tau}\leq\sum_{\tau=t}^{t+\Delta}f\leq\Delta\cdot f, \tag{3}$$

and

$$\sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}C\cdot z\leq B\cdot z, \tag{4}$$

it can be known

$$\begin{aligned}\sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}C(\mathbf{P}_n) &\leq \sum_{t=1}^{T-\Delta}\sum_{m=1}^{M}\left(y_{mt}\cdot\sum_{\tau=t}^{t+\Delta}\sum_{\mathbf{P}_n\in\mathcal{P}}\mathbf{P}_n^{m\tau}\right)-\sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}C\cdot z\\ &\leq \sum_{t=1}^{T-\Delta}\sum_{m=1}^{M}(y_{mt}\cdot\Delta\cdot f)-\sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}C\cdot z\\ &= \Delta\cdot f\cdot\sum_{t=1}^{T-\Delta}\sum_{m=1}^{M}y_{mt}-\sum_{\mathbf{P}_n\in\widehat{\mathcal{P}}}C\cdot z\\ &\leq \Delta\cdot f\cdot\sum_{t=1}^{T-\Delta}\sum_{m=1}^{M}y_{mt}-B\cdot z\\ &\leq \Delta\cdot f\cdot C(OPT_f)\\ &\leq \Delta\cdot f\cdot C(OPT).\end{aligned} \tag{5}$$

Appendix B
Benchmark Algorithms

Definition 1* (Weak Dominance)*: For two trajectory sets $\mathcal{R}_q$ and $\mathcal{R}_q'$, if $f_1(\mathcal{R}_q')\geq f_1(\mathcal{R}_q)$ and $f_2(\mathcal{R}_q')\geq f_2(\mathcal{R}_q)$ hold simultaneously, we define trajectory set $\mathcal{R}_q'$ to be weakly dominant over trajectory set $\mathcal{R}_q$ (denoted as $\mathcal{R}_q'\succ=\mathcal{R}_q$).

We adopt insertion and swap mutation operations. The insertion operation helps quickly increase the coverage of PoIs under the budget constraint, while the swap operation improves the diversity of solutions. The detailed procedures are presented in Algorithms 1 and 2.

We also transform the periodic wildlife population monitoring problem under budget constraints into a set cover problem. First, we relax the problem by assuming that each PoI needs to be visited at regular intervals of Δ time periods, starting

Algorithm 1 EA Periodic Data Collection (EA-PDC)

Input: All items $\mathcal{M} = \{1, 2, \ldots, M\}$, $\mathcal{P}$
Output: $\widehat{\mathcal{P}}$

Let $\widehat{\mathcal{P}} \leftarrow \emptyset$
Let $\mathcal{C} = \{\mathcal{R}_1, \ldots, \mathcal{R}_q, \ldots, \mathcal{R}_Q\}$, $1 \leq q \leq Q$ and for each vehicle trajectory collection $\mathcal{R}_q \leftarrow \emptyset$
while the stopping criterion is not satisfied **do**
  Randomly select a vehicle trajectory collection $\mathcal{R}_q$ from population $\mathcal{C}$
  Generate a new vehicle trajectory collection $\mathcal{R}'_q$ by applying mutation
  if $\mathcal{R}'_q \succ= \mathcal{R}_q$ and $\mathcal{R}'_q \notin \mathcal{C}$ **then**
    $\mathcal{C} = (\mathcal{C} \setminus \{\mathcal{R}_q\}) \cup \{\mathcal{R}'_q\}$
  end if
end while
$\mathcal{R} = \arg\max_{\mathcal{R} \in \mathcal{C}}(f_1(\mathcal{R}), f_2(\mathcal{R}))$
$\widehat{\mathcal{P}} = \mathcal{R}$
return $\widehat{\mathcal{P}}$

Algorithm 2 Mutation

Input: A vehicle trajectory collection $\mathcal{R}_q$
Output: A new vehicle trajectory collection $\mathcal{R}'_q$

Let $s = 0$, $r = Poisson\, distribute\, \text{random}(\lambda = 1)$
while $s < r$ **do**
  if $\mathcal{R}_q$ is the initial vehicle trajectory collection **then**
    $\mathcal{R}'_q = \text{Insertion}(\mathcal{R}_q)$
  else
    if $random(0, 1) < 0.5$ **then**
      $\mathcal{R}'_q = \text{Insertion}(\mathcal{R}_q)$
    else
      $\mathcal{R}'_q = \text{Swap}(\mathcal{R}_q)$
    end if
  end if
  $s = s + 1$
end while
return $\mathcal{R}'_q$

from time instant t. However, the relaxed solution may not always be feasible for the original problem, as it could lead to visit intervals greater than Δ. To address this, for PoI m, it must be visited at least $\lceil \frac{T-t}{\Delta} \rceil$ times within $(t, \mathcal{T})$. We define this as $\lceil \frac{T-t}{\Delta} \rceil$ elements, and with M PoIs, the set contains a total of $M \cdot \lceil \frac{T-t}{\Delta} \rceil$ elements. We use the set $\mathcal{U}$ to represent this collection, based on which we design the Greedy-PDCA algorithm. Where $G(P)$ denotes the set of elements in $\mathcal{U}$ that can be covered by P, and $\mathcal{A}$ represents the set of elements that have already been selected.

Algorithm 3 Greedy Periodic Data Collection Algorithm (Greedy-PDCA)

Input: Δ, t, $\mathcal{T}$, $\mathcal{P}$, $\mathcal{M}$
Output: $\widehat{\mathcal{P}}$

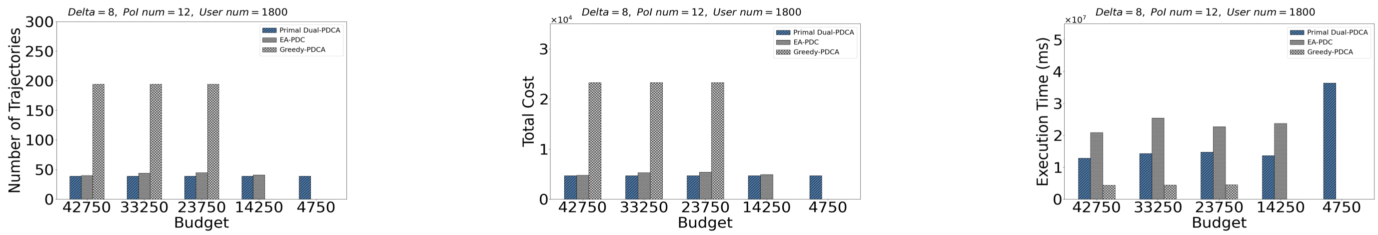
Let $\widehat{\mathcal{P}} \leftarrow \emptyset$
for $1 \leq t \leq \Delta$ **do**
  Construct a collection $\mathcal{U}$ according to the Δ, t, $\mathcal{T}$, $\mathcal{P}$, $\mathcal{M}$
  $\mathcal{A} \leftarrow \emptyset$
  while $\mathcal{A} \neq \mathcal{U}$ **do**
    Find the most cost-effective trajectory from set $\mathcal{P}$ in the current iteration, say P
    Let $\alpha = \frac{V(P)-C}{|G(P)-\mathcal{A}|}$
    Pick P into $\widehat{\mathcal{P}}_t$, and for each element $e \in G(P) - \mathcal{A}$, set cost $c(e) = \alpha$
    $\mathcal{A} \leftarrow \mathcal{A} \cup G(P)$
  end while
  $\widehat{\mathcal{P}} = \widehat{\mathcal{P}} \cup \widehat{\mathcal{P}}_t$
end for
return $\widehat{\mathcal{P}}$

APPENDIX C
IMPACT OF THE BUDGET (SMALLER IS BETTER)

This experiment aims to systematically evaluate the performance differences of Primal Dual-PDCA, EA-PDC, and Greedy-PDCA under different budget levels (42750, 33250, 23750, 14250, and 4750) in terms of data value and algorithm execution time. Specifically, under a monitoring period of 8 days, the algorithms select a subset of trajectories from 1800 feasible mobile vehicle trajectories for 12 PoIs, aiming to satisfy the periodic visiting requirements while minimizing the total cost under the given budget constraint.

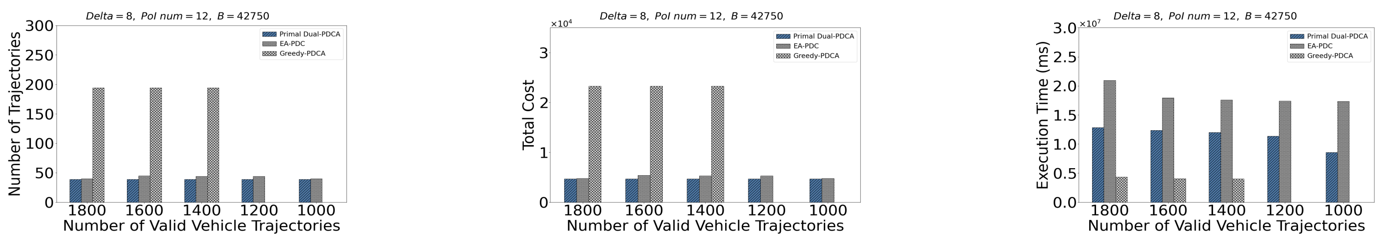
Figure 1(a) shows the number of selected trajectories generated by each algorithm under different budget levels. As the budget gradually decreases from 42750 to 4750, the difficulty of generating a feasible trajectory set increases significantly. In particular, Greedy-PDCA fails to generate a feasible trajectory set when the budget is no higher than 14250. EA-PDC cannot stably produce feasible solutions when the budget is 4750, and the generated solutions may violate the budget constraint. These observations indicate that the budget constraint has a significant impact on the trajectory selection process. In contrast, Primal Dual-PDCA is able to generate feasible solutions under all budget settings, demonstrating that the primal–dual based approach exhibits stronger feasibility in extremely budget-constrained scenarios.

Figure 1(b) presents the comparison of the total cost achieved by different algorithms under various budget levels. It can be observed that under the same budget condition, except for cases where feasible solutions cannot be stably generated, Greedy-PDCA consistently incurs significantly higher total costs than the other two algorithms. This is mainly because it selects a larger number of trajectories, leading to a rapid accumulation of overall cost. EA-PDC achieves a moderate level of total cost, whereas Primal Dual-PDCA consistently obtains the lowest total cost across all budget settings. Notably, even under tight budget constraints, Primal Dual-PDCA is still able to maintain a relatively low total cost while producing feasible solutions. This further demonstrates its effectiveness and stability in handling budget-constrained optimization prob-



(a) Number (b) Cost (c) Execution time

Fig. 1: Performance of the three algorithms under five different budget for meeting data collection requirements.



(a) Number (b) Cost (c) Execution time

Fig. 2: Performance of the three algorithms under five different number of vehicles for meeting data collection requirements.

lems. The results indicate that the algorithm can better balance trajectory utility and cost while satisfying periodic coverage requirements.

Figure 1(c) compares the execution time of the three algorithms under different budget conditions. Overall, when feasible solutions can be generated, Greedy-PDCA consistently achieves the shortest execution time. This is mainly due to its simple algorithmic structure and greedy decision-making strategy, which leads to relatively low computational complexity. EA-PDC requires longer execution time because the evolutionary mechanism involves multiple iterations and population updates, resulting in additional computational overhead. Primal Dual-PDCA achieves execution time between the two methods and remains relatively stable across most budget settings. However, under the lowest budget condition (4750), the execution time of Primal Dual-PDCA slightly increases. This is because extremely tight budget constraints require more dual adjustments to generate all feasible solutions, thereby increasing the computational cost.

Appendix D
Impact of the Number of Vehicles (Smaller is Better)

This experiment aims to systematically evaluate the performance differences of Primal Dual-PDCA, EA-PDC, and Greedy-PDCA under different numbers of available vehicle trajectories (1800, 1600, 1400, 1200, and 1000) in terms of data value and algorithm execution time. Specifically, under a monitoring period of 8 days and a total budget of 42750, the algorithms select a subset of trajectories from the available vehicle trajectories to serve 12 PoIs, aiming to satisfy the periodic visiting requirements while minimizing the total cost under the budget constraint.

Figure 2(a) illustrates the variation in the number of selected trajectories as the number of available vehicle trajectories gradually decreases. It can be observed that Greedy-PDCA still selects a relatively large number of trajectories when sufficient trajectories are available (e.g., 1800–1400). This is mainly because the algorithm introduces trajectories based on a local greedy criterion, prioritizing those with the highest immediate benefit while lacking a global mechanism to control trajectory redundancy. As a result, it tends to include an excessive number of trajectories. When the number of available trajectories is insufficient (e.g., 1200–1000), Greedy-PDCA fails to generate a feasible trajectory set that satisfies all constraints. In contrast, EA-PDC, whose results are averaged over multiple independent runs, and Primal Dual-PDCA are able to consistently generate feasible solutions under different scales of available trajectories. This indicates that both algorithms exhibit stronger trajectory selection capability, while Primal Dual-PDCA outperforms EA-PDC in terms of trajectory efficiency.

Figure 2(b) presents the comparison of the total cost achieved by different algorithms under the same experimental settings. The variation in total cost closely follows the trend observed in Figure 2(a), as the total cost generally correlates with the number of selected trajectories.

Figure 2(c) compares the execution time of the three algorithms under different numbers of available vehicle trajectories. Overall, as the number of candidate trajectories decreases, the execution time shows a decreasing trend because the reduction in candidate trajectories directly shrinks the search space of the algorithms. When feasible solutions can be generated, Greedy-PDCA consistently achieves the shortest execution time, mainly because it relies only on simple greedy rules for decision-making and does not require complex iterative updates or global search. EA-PDC exhibits significantly longer execution time than Primal Dual-PDCA, since the evolutionary algorithm involves multiple generations of iterations, as well as crossover and mutation operations, whose computational cost is closely related to the number of iterations. As a result, its execution time tends to remain relatively stable even when the number of candidate trajectories decreases. Primal Dual-PDCA achieves execution time between the two methods and shows a steady decrease as the number of available trajectories decreases. This indicates that while incorporating the dual-update mechanism, the algorithm still maintains good computational efficiency and achieves a reasonable balance between solution quality and computational overhead.